Crop Recommendation and Yield Prediction using Machine learning and Deep learning models

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***Abstract*—Agriculture remains the predominant livelihood provider in India contributing to India's Gross Domestic Product (GDP). Recent advancements in technology like precision agriculture and agri-tech solutions, have improved efficiency. However, the domain faces challenges due to climate change, water scarcity, and land degradation. This research focuses on leveraging machine learning (ML) and deep learning (DL) models to address these challenges through crop recommendation and yield prediction. In this paper a combined system for crop recommendation and yield prediction is done using a comprehensive dataset that has factors like pH, rainfall, state name, crop type, crop name, N, P, K, humidity, area in hectares, production, temperature, and yield in tons per hectare. Preprocessing techniques were applied to improve data quality and model accuracy. The dataset is trained using machine learning models (K-Nearest Neighbors, Random Forest and Decision Tree) and deep learning models (Artificial Neural Network-ANN, Long Short-Term Memory-LSTM, Deep Neural Network-DNN). The findings indicate that K-Nearest Neighbors (KNN) and Random Forest among the other ML models provide superior accuracy. Random Forest (RF) achieved the best accuracy in crop recommendation (99.3%) and KNN performed well in yield prediction (R^2 score: 0.985). Among DL models, LSTM has higher efficiency both in terms of crop recommendation (97%) and yield prediction (R^2 score: 0.930). Integration of these models will help farmers to make decisions related to crop selection and the yield which will eventually result in increased productivity and sustainability.The insights from this paper highlights the advantages of data driven approaches in handling difficulties encountered by farmers, thus paving way for a sustainable and food secure future.**

Keywords — crop recommendation, yield prediction, machine learning, deep learning

I. INTRODUCTION

Agriculture, with its associated sectors, provides the major source of livelihood for many people in India, especially in the rural parts. It is one of the major contributors to GDP(Gross Domestic Product) of India. Agriculture in recent years has witnessed both progress and challenges. Technological advancements like precision agriculture and agri-tech solutions boost efficiency and sustainability, while concerns about climate change, water scarcity, and land degradation loom large. Meanwhile, rising demand for plant-based alternatives is reshaping agricultural practices. The future hinges on addressing these challenges, harnessing technological potential, and promoting sustainable practices to ensure a food-secure future for all. Within this dynamic landscape, crop recommendation and yield prediction have emerged as vital tools for farmers navigating the complexities.

Precision agriculture employs sensors and drones, weaving a tapestry of field-specific data on soil health, weather patterns, and pest presence. These insights empower tailored recommendations for crop selection, fertilizer application, and irrigation, maximizing both yield and resource use. Machine learning algorithms, fueled by vast historical datasets, are pushing the boundaries of yield prediction and crop recommendation. Farmers can leverage these insights to optimize resource allocation and maximize harvests, making informed decisions amidst unpredictable seasons. By analyzing soil properties and weather patterns, recommendations can favor crops less vulnerable to water scarcity or requiring fewer chemical inputs. Yield predictions can be recalibrated to consider sustainable practices, allowing farmers to assess trade-offs between yield and environmental impact. This synergy between technology and sustainability paves the way for a future where productivity flourishes alongside environmental responsibility. However, the road is not without challenges. Climate change throws a curveball, presenting unpredictable extreme weather events that current models struggle to fully capture. Ongoing research and adaptation are crucial to refine predictive abilities and ensure recommendations remain relevant in a changing climate. Similarly, water scarcity demands real-time data-driven optimization of irrigation practices, while crop recommendations need to favor drought-resistant options where needed. Moreover, integrating these tools within broader agricultural policies and support systems will ensure equitable access and empower farmers to navigate the uncertainties of the future. By embracing these advancements while addressing the challenges, crop recommendation and yield prediction can be powerful allies in building a more resilient, sustainable, and food-secure agricultural future.

Recent advancements in data-driven approaches and machine learning are transforming the agricultural landscape, with the potential to improve efficiency, accuracy, and sustainability across various domains. Machine learning models are being trained on diverse datasets encompassing weather patterns,

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soil health, and management practices to predict crop yields. This foresight empowers farmers to make informed decisions regarding allocation of resources, storage, and market opportunities.

The research contributes to the existing efforts taken to address the difficulties faced by farmers by using data driven approaches such as machine learning and deep learning techniques.Particularly the research focuses on developing a combined system for both crop recommendations and crop yield prediction with the help of a comprehensive dataset that has factors like pH, rainfall, state name, crop type, crop name, N, P, K, humidity, area in hectares, production, temperature, and yield in tons per hectare.The study applies ML models like KNN, RF, and DT and DL models like (ANN ,LSTM, DNN) to improve the accuracy of crop recommendation and crop yield prediction.This research intends to help farmers make informed decisions which contributes to increased productivity and resilience in unexpected environmental challenges.

II. LITERATURE REVIEW

Sharma et al. (2023) deals with yield prediction comparing five ML and Dl algorithms like random forest, decision tree, XGBoost, CNN and LSTM [1]. CNN was found to provide better performance. The dataset consist of parameters like state name, district name, season, crop name, rainfall, wind speed, area, area under irrigation, production, yield, humidity and crop year.

Hasan et al. (2023) deals with crop recommendation using KRR (K-nearest Neighbor Random Forest Ridge Regression) [2].The dataset consists of 5 categories of crops each having 1400 samples of Bangladesh. The parameters include rainfall, sunshine, wind speed ,humidity,cloud coverage,production, area, production amount, maximum temperature and minimum temperature.

Garg et al. (2023) deals with Wrapper-Part-Grid method [3]to classify data to provide crop recommendation. In Wrapper -PART-Grid approach groups grid search method for hyperparameter optimization, wrapper selection strategy and partial decision tree classifier (C4.5).The dataset has 2200 records and parameters like N,P,K,pH,temperature ,humidity and rainfall. By using grid optimization and part algorithm 99.31% accuracy is obtained outperforming other models like instance based learning with k parameter ,reduced error pruning tree(REP),multilayer perceptron (MLP) and CDT.

Thilakarathne et al. (2022) Compares predictive Machine Learning algorithms like KNN, RF , DT , XGBoost and SVM[4] to find out the best Machine Learning Algorithm for crop recommendation as cloud based service.The findings indicate that Random Forest has the best score(97.18%). making it the best algorithm for the proposed purpose.This study highlights the effectiveness of the Random Forest algorithm handling huge dataset and identifying complex underlying patterns.

Gosai et al. (2021) deals with recommending the appropriate crop using factors like N,P,K,temperature,soil,amount of rainfall and humidity .Comparing different ML algorithms like SVM, Naive Bayes, DT, XGBoost,logistic regression and Random Forest [5] this paper concludes that XGBoost provides high accuracy.

Alagurajan et al. (2020) used the dataset consists of crops like rice,wheat and potato which are grown in Chennai with parameters like rainfall,humidity,urea soil and temperature. Yield prediction is based on machine learning algorithms like ANN,Decision forest and SVM and ANN [6] is found to have more accuracy. The study emphasized that ANN is capable of learning from huge datasets and adapt to several input conditions making it preferable for crop yield prediction.

Van Klompenburg et al. (2020) gives a comprehensive overview on the importance of machine learning in the field of crop yield prediction and gives details of commonly used algorithms like ANN, CNN, LSTM, DNN [7] for deep learning approaches.The study reviewed and analyzed 50 selected research papers from a collection of 567 studies obtained from six electronic database. study reveals that the commonly used feature in the deep learning models were temperature,soil type and amount of rainfall. also highlights that ANN is commonly used for yield prediction.

Elavarasan et al.(2020) grouping the strengths of DL(deep learning) and RL(reinforcement learning). By constructing a Deep Recurrent Q-Network model [8], They have successfully mapped raw data to crop yield values. This model leverages Recurrent Neural Networks and Q-Learning to build a comprehensive framework for crop yield prediction, achieving an impressive accuracy of 93.7%.

Reddy et al. (2019) deals with proposing an ensemble model to recommend suitable crops. Algorithms like CHAID, NaiveBayes, Randomtree, K-Nearest Neighbor were used [9] and proposed an ensemble model. The ensembled model employs majority voting techniques to recommend crop based on soil. The classified images produced by this method includes parameters like crop yield,state,district wise crops, weather which are used for yield prediction.

Bondre et al. (2019) deals with ML techniques to predict the crop yield, classify soil and recommend fertilizer, SVM and Random Forest models were used for this purpose [10] and concluded that SVM produced 99.47% accuracy more than Random Forest in case of crop yield.

Oikonomidis et al. (2022)used DL (deep learning)models [11] to predict yield using Deep Neural Networks, XGBoost, Convolutional Neural Networks, CNN-XGBoost, CNN-Long Short-Term Memory and, CNN-Recurrent Neural Networks on a soybean dataset. A hybrid CNN-DLL model provided better accuracy than others with RMSE(0.266) and R^2(0.87).

Jhajharia et al. (2023) developed machine learning models (ML) topredict crop yield [12]for the state of Rajasthan for

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five major crops using Random Forest, LSTM, Gradient Descent, SVM and Lasso regression. Random forest performed the best ,achieving an RMSE of 0.035, an R^2 of 0.963 and MAE of 0.0251.

Nidhi H Kulkarni et al. (2018) developed a crop recommendation system [13] that utilizes ensembling techniques which enhances crop productivity accurately and suggests crops based on characteristics such as soil and climatic conditions. Achieved an accuracy of 99.91%

Sandhya et al. (2022) developed a Crop Recommendation System [14] which utilizes ensembling techniques that suggests the suitable crop based on characteristics such as soil and environmental parameters. KNN, Random Forest, Naïve Bayes, Logistic Regression, and SVM are the algorithms used by them to develop a crop recommendation system.

III. PROPOSED METHODOLOGY

We have proposed a methodology using machine learning models such as KNN, Random Forest, DT and DL models such as DNN, ANN, LSTM to recommend the suitable crop and predict the yield of the crop.

A. Dataset description

We have combined the Crop recommendation dataset and Crop yield dataset to get a comprehensive dataset with 12 columns such as: state name, crop type, crop name,nitrogen, phosphorous, potassium, humidity, rainfall, area (hectares),temperature, pH, production and yield in tons per hectare.

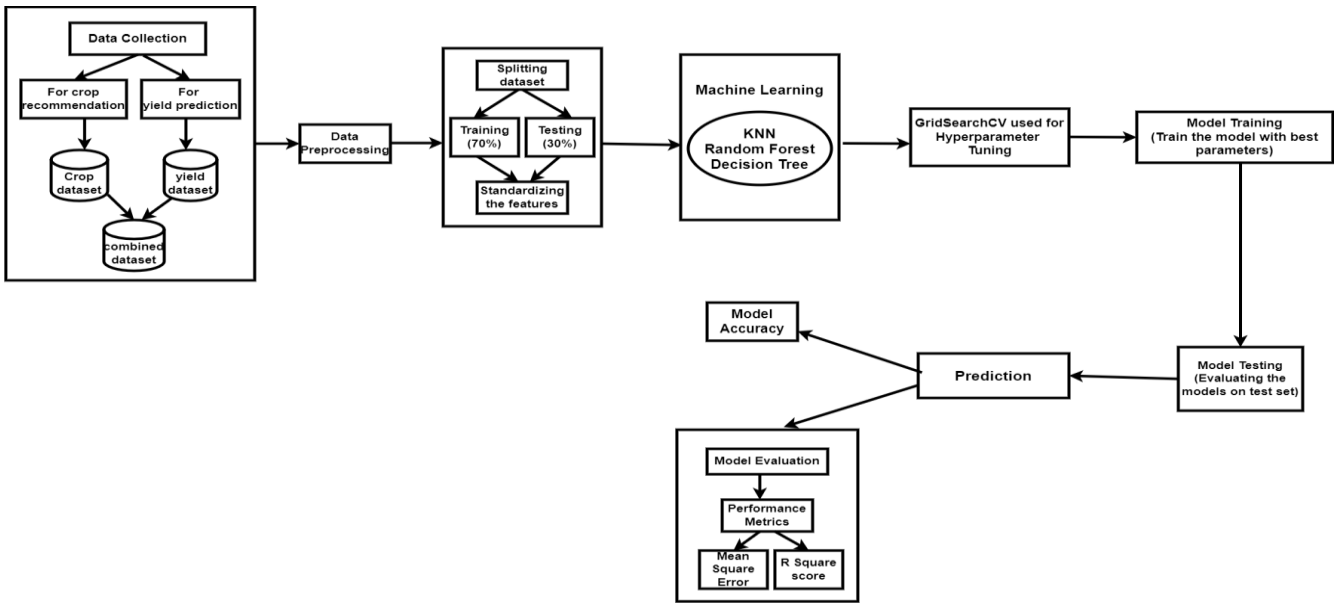

Fig. 1. Block Diagram for Machine learning models

B. Data preprocessing

The final dataset is obtained by performing certain preprocessing techniques such as handling null values, data type conversion and label encoding. These steps help improve model accuracy, enable dataset quality enhancement and prevent overfitting.

C. Train test split

Three different ML and DL models were trained using the dataset. The performances of the models were compared and evaluated.The models used are Decision Tree, KNN,Random Forest,ANN,DNN and LSTM. The dataset is divided into two parts as a training set (70%) and testing (30%) . In the case of ML, the training set constitutes 80% and the testing set constitutes 20% for deep learning. Then, the data is scaled using the Standard Scaler. Standardizing the features in testing and training dataset helps the model to improve its performance.

D. Model training platform

Google Colab is a free platform primarily used for python programming, especially for data processing and machine learning tasks. It runs on cloud and offloads computing work on local computers. In addition, the device utilized for dataset acquisition, loading and preprocessing, model training is equipped with a I3 processor, 64-bit operating system and 8GB of RAM, which effectively meets the model training requirements mentioned in this study.

E. Machine Learning models used

a) KNN:

KNN is a supervised learning model that is used for classification and regression. For classification, it allocates a label to newly derived data using the distance between the older data and newly derived data. For numerical regression, it finds the K nearest data points of a given input and finds the mean of their target values. K refers to the number of neighbors to be considered. Cross validation is used to find

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the best k value using GridSearchCV based on the accuracy metric. The base model with k value as 1 in the case of crop recommendation and k value of 2 in the case of crop yield prediction are found to have higher accuracy from cross validation. Then, best models are trained on scaled x_train and y_train and predictions are obtained based on the scaled x_test data.

b) Random Forest:

Random forest is a combination of multiple decision trees which is used for both classification and prediction. For classification, it predicts new data by combining the predictions of the tree. For regression, it specifies the mean of the tree. Cross validation creates a GridSearchCV to find the best hyperparameter combination and used to find the value of n_estimators as 200 for recommendation and n_estimators as 250 for yield prediction. Best parameters are used to train the model and predictions are obtained based on the scaled x_test data.

c) Decision Tree:

A DT is a prophetic machine learning model which partitions the data recursively into subsets based on its features to create a tree structure where nodes denote a decision based the value of the features. For crop recommendation, criterion as entropy, random_state as 2, and max_depth as 5 are used as hyperparameters.

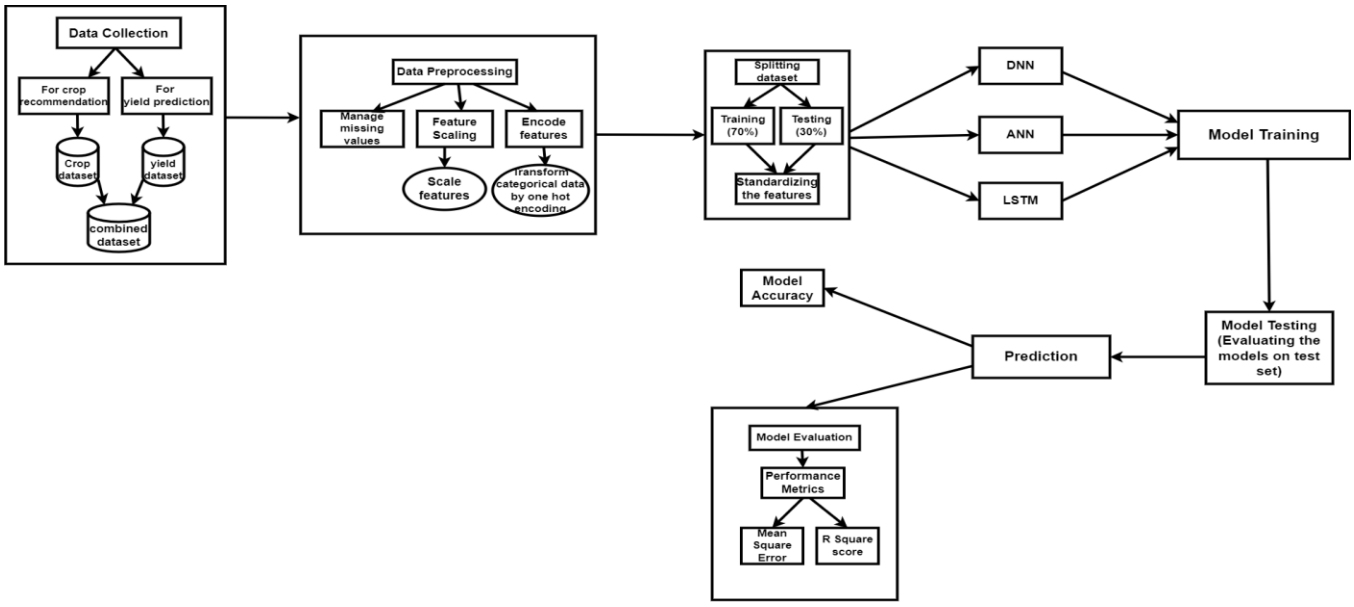

Fig. 2. Block Diagram for Deep learning models

F. Deep Learning models used

d) DNN:

Deep neural networks increase the concealed layers between input and output layers which are used frequently for their accuracy. For recommendation the network has an input layer matching of features with two conceal layers and a resultant layer with a soft_max activation function to recommend crop based on the given inputs. For prediction the network includes two hidden layers as input followed by a resultant layer with a single neuron to predict the continuous yield value. The evaluation is based on the accuracy after compiling with Adam optimizer.

e) ANN:

ANN is a significant deep learning model. It has three layers such as input layer, output layer and conceal layer. We have used two conceal layers with 64 neurons each using the ReLu activation function. It enables the model to learn complex patterns from the dataset. The model is checked with Adam optimizer, sparse categorical cross-entropy as loss function and suitable for multiclass diversion, and accuracy as metric for training and evaluation.

f) LSTM:

LSTM is a deep learning technique, a kind of recurrent neural network(RNN) which aims to deal with the vanishing gradient problem. For recommendation, the input layer has a single LSTM layer followed by a dense output layer. It has a softmax activation function and the number of neurons in the output layer is respect to the number of unique crop classes, which is determined by the label encoder. For prediction, the input consists of an LSTM layer which has 64 units and ReLU activation, followed by a dense layer with 64 neurons and another ReLU activation. The output layer has a single neuron to predict the continuous yield value. The evaluation is based on the accuracy after compiling with Adam optimizer.

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IV. Result and Discussion

A. Model Evaluation Metrics

The metrics used for yield prediction is r2 score, whereas accuracy is used to evaluate the models for crop recommendation.

a) Accuracy: The ratio between correctly predicted instances and the total instances.

$$Accuracy = (True\ Positives + True\ Negatives)\ /\ Total\ instances \quad (1)$$

b) R2-score: The coefficient of determination, measures the ability of the regression line to approximate the actual data.

$$R^2 = 1 - (SS_{res} / SS_{tot}) \quad (2)$$

B. Model Comparison

TABLE I. Evaluation metrics

Model	Accuracy	R2 score
KNN	96.8	0.985
Random Forest	99.3	0.980
Decision Tree	89.4	0.959
DNN	94.8	0.941
ANN	95.2	0.941
LSTM	97.0	0.930

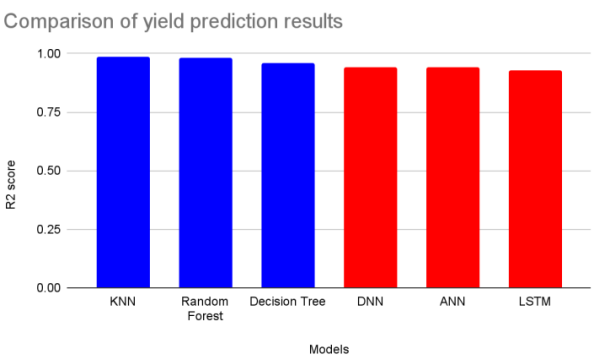

Fig. 3. Yield prediction results

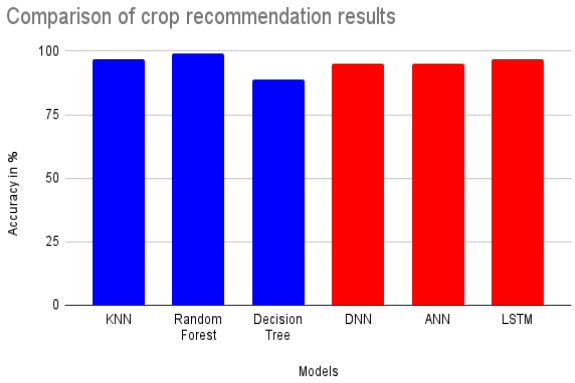

Fig. 4. Crop recommendation results

C. Comparison with existing works

Our work resulted in higher accuracies and R2-Scores than previous work done in this field. Thilakarathne et al. in their work [4] achieved accuracies - 97.18 for Random Forest, 96.36 for KNN, and 86.64 for Decision Tree. Here, we have achieved better accuracies - 99.3 for Random Forest, 96.80 for KNN, and 89.40 for Decision Tree. Similarly for yield prediction, Oikonomidis et al. in their paper [11] gained R2-Scores such as 0.87 for CNN-DNN hybrid model and 0.67 for CNN-LSTM whereas, our models resulted in 0.941 for DNN and 0.93 for LSTM. Jhajharia, K et al. [12] showed R2-Scores of 0.96 for Random Forest, 0.75 for LSTM and we got an increased R2-Score 0.98 for Random Forest and 0.93 for LSTM.

V. Conclusion and Future Work

This paper highlights the potential of utilizing machine learning (ML) techniques and deep learning (DL) techniques for crop recommendation and yield prediction in advanced precise agriculture. Dataset is obtained by combining dataset from various resources that enables robust model training and evaluation. By combining datasets from various sources, we ensure robust model training and evaluation. These integrated datasets provide a comprehensive view, capturing diverse factors affecting crop growth and yield.

The findings indicate that the K-Nearest Neighbors model (KNN) and Random Forest model amongst other ML models provide superior accuracy. Random Forest yields the highest accuracy in recommendation of crops (99.3%) and KNN model performing well in yield prediction (R^2 score: 0.985). In terms of DL models , LSTM has higher efficiency both in terms of crop recommendation(97%) and yield prediction(R^2 score: 0.930).

Integration of these models into agricultural practices offers significant benefits. Farmers can make use of these techniques to make decisions about selection of crop and yield prediction,consequently enhancing productivity and sustainability.

Our future work aims to predict the price for each crop to estimate the approximate price to achieve more profit. We also aim to utilize the best performing models in a user interface for users to easily get recommendations and predictions based on the input provided.Furthermore,

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incorporating weather prediction models can help in planning irrigation schedules and optimizing resource usage, leading to more sustainable farming practices. Our future works could also include developing comprehensive dashboards that provide actionable insights, enabling farmers to carry outactions with greater confidence and precision.

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